



Web system-assisted ratiometric fluorescent probe embedded with machine learning for intelligent detection of pefloxacin

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ABSTRACT

It is urgent to realize the quantitative detection of pefloxacin (PFLX), which has been misused to accumulate in the environment and animals. The traditional detection method is not only time-consuming and laborious, but also can only detect a single sample at a time. In this research, a BSA-AuNCs-GMP-Tb ratiometric fluorescent nanocomposite probe with red fluorescence in the initial state was constructed by in situ preparation of guanine nucleotide-terbium ion (GMP-Tb) complex using bovine serum albumin-coated gold nanoclusters (BSA-AuNCs) as the substrate material. Under the optimal experimental conditions, the nanoprobe had a wide visual detection range (0.01 μM –60 μM) with an ultra-low detection limit (5.37 nM) for PFLX. A paper-based visualized sensor was further designed with a smartphone color recognition app to achieve qualitative and semi-quantitative detection of PFLX. In addition, merging You Only Look Once version 5 (YOLO v5) target detection algorithm and machine learning regression algorithm, a web-based PFLX concentration prediction system was developed to achieve the quantitative detection of PFLX in actual environment samples. The system was developed to achieve batch detection of PFLX in environmental samples, which greatly improved the speed of PFLX detection.

1. Introduction

Pefloxacin (PFLX), a third-generation quinolone antibiotic, has been widely used in treating viral and bacterial infections due to its ability to inhibit DNA synthesis [1]. In addition, PFLX is also used to treat various infectious diseases (such as urinary system, respiratory tract) and bacterial infections in livestock [2,3]. However, the excellent drug properties, few side effects, and low price of PFLX make it illegal to be used as a growth-promoting agent for poultry [4]. The abuse and incorrect treatment of PFLX accumulate in the environment due to its poor degradability and high water solubility [5,6]. Residues of PFLX in animals, wastewater, and soil have brought significant health problems to humans [7]. Therefore, there is an urgent need to develop a sensitive and rapid method for detecting PFLX residues in the actual environment (such as food, water environment, etc.) [8].

At present, various techniques have been used to detect PFLX, such as chemiluminescence [9], capillary electrophoresis [10], fluorescence method [5,11], high-performance liquid chromatography [12], and longitudinal surface plasmon resonance [13]. Compared with other

detection methods, the fluorescence method is considered to be a promising method for detecting antibiotic residues because of its simple operation, strong selectivity, high sensitivity, fast response, and good visualization [14–16]. Compared with the traditional single-signal fluorescent nanoprobe, the detection accuracy of the ratiometric fluorescent probe is not easily affected by the probe concentration, instrumental parameters, solution polarity, and other factors and has high interference tolerance [17–19].

It is well known that lanthanide ions have unique photophysical properties and narrow emission bands [20]. Lanthanide-based coordination polymers can be used as fluorescent probes for small molecules, anions, and metal ions sensing ascribed the structural diversity, inherent biocompatibility, excellent encapsulation ability, structural diversity, and sharp linear and complex-independent emission bands [21–23]. Zhou et al., [24] showed that nucleotides are templates for synthesizing many functional nanomaterials. Lanthanide ion-nucleotide coordination polymers have unique luminescent properties, such as long fluorescence lifetime, high photochemical stability, and sharp emission [25]. It has been reported that the infinite coordination polymer (ICP) based on

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guanosine monophosphate (GMP) and lanthanide terbium ion (Tb^{3+}) has the characteristics of high structural flexibility and fast response [26]. There is a strong antenna effect between GMP and Tb^{3+} , which can effectively sensitize Tb^{3+} and promote its luminescence [20,27].

As a new type of fluorescent material, gold nanoclusters (AuNCs) have the characteristics of simple preparation, simple functionalization, low toxicity, significant Stokes shift, emission mobility, and long decay life, especially aggregation-induced emission (AIE), which makes AuNCs widely used in a variety of fluorescent nanomaterials [28–30]. The reduction of Au with suitable ligands or the use of templates has been widely used in the synthesis of AuNCs [31]. Among them, AuNCs encapsulated with bovine serum albumin (BSA) molecular ligands have better biocompatibility and higher stability [32].

With the development of smartphones, camera pixels continue to increase, and more and more data is stored in the form of images. Deep learning (DL) is widely used to process image data. In 2015, Joseph and Girshick et al., [33] proposed a target detection model called YOLO (You Only Look Once). YOLO is a single-stage object detection method based on DL. The algorithm treats bounding boxes and classification as a regression problem [34,35], and it can directly obtain the target category and coordinate information [36]. Lu et al., [37,38] developed a WeChat applet based on the YOLO v3 target detection algorithm. Combined with the "on-off-on" mechanism of the fluorescent probe (Fe-MIL-88 NH_2 /Au NCs), the quantitative detection of Hg^{2+} and thiram was realized. Then, the detection of Cu^{2+} was realized by combining the ratio fluorescent probe based on three-color quantum dots. Machine learning (ML) is to find the relevant rules (such as functional relationships) that conform to the data set from the known data set and use this rule to predict the unknown data. ML algorithms have been widely used to detect antibiotics [39,40], antibiotic resistance prediction [41] and so on. With the development of "Internet +", smart devices are integrated into many traditional industries. Django is a Python-based Web framework that can achieve the effect of mobile websites through some internal network configuration and external network mapping operations [42]. The framework is widely used because of its lightweight, front-end and back-end separation, and short development cycle [43].

As an significant part of artificial intelligence (AI), DL has broad and

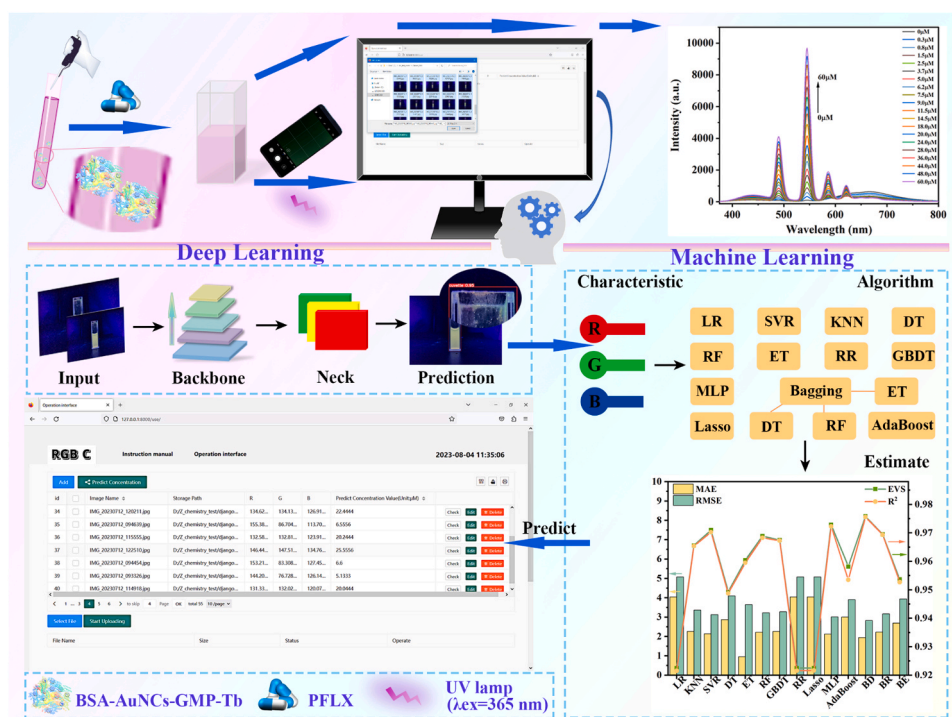
far-reaching significance in many studies. DL algorithms mainly perform classification and prediction tasks. The single-stage target detection algorithm based on YOLO v5 model has the characteristics of high efficiency, high detection accuracy and strong anti-interference ability. It provides technical support for object recognition and small target recognition in complex background environment. Furthermore, the method of detecting PFLX based on machine learning-assisted ratio-metric fluorescence sensor has not been reported. In this research, a ratio-metric fluorescent nanoprobe based on the integration of GMP-Tb and BSA-AuNCs was designed. Based on fluorescence changes, a paper-based sensor and a Django-based Web system were developed for visual, convenient, and online detection of PFLX. Among them, due to the sensitization between PFLX and Tb^{3+} , the characteristic peak of Tb^{3+} increased, and the fluorescence intensity peak of BSA-AuNCs-GMP-Tb tended to be stable with the increase of PFLX concentration so that the fluorescent probe showed a red to green change. A paper-based sensor was prepared to achieve rapid visual detection of PFLX by fixing the nanocomposite probe on the filter paper. Based on the Python language, an interactive PFLX concentration prediction system (Scheme 1) was developed, embedded with ML algorithms. The YOLO v5 algorithm based on DL can identify the target and extract the color information of the mixed solution fluorescence image based on the fluorescent probe uploaded by the user, which provided reliable data for predicting PFLX concentration. In addition, this study also used twelve ML algorithms for model training. The best algorithm for quantitative detection of PFLX was finally determined and embedded into the system through model evaluation.

2. Experimental section

The Supporting Information listed the synthesis methods of reagents, experimental instruments, and construction of concentration prediction system based on web involved in this study.

2.1. Fluorescence detection of pefloxacin

70 μL of BSA-AuNCs-GMP-Tb probe stock solution was taken in a



Scheme 1. Concentration prediction system embedded with ML algorithm assisted ratio-metric fluorescent probe to detect PFLX.

glass cuvette, and 1930 μL of HEPES buffer (10 mM, pH=4) was added to form a stable system of 2 ML. An equal amount of PFLX solution was added dropwise to the stabilized system, and the fluorescence spectrum under 365 nm excitation light was recorded, and the fluorescence pictures under 365 nm UV lamp were taken using the camera function of the smart device.

2.2. Preparation of visual detection test paper and detection of pefloxacin

The ordinary filter paper without background interference was cut and immersed in a mixed 800 μL of BSA-AuNCs-GMP-Tb probe stock solution and 800 μL of HEPES buffer (10 mM, pH=4). After soaking for 3 h, it was dried at room temperature to obtain a visual detection test paper (paper-based sensor). Different amounts of PFLX were added to the test paper, and the fluorescence changes under the 365 nm UV lamp were recorded with the smartphone's camera.

2.3. Detection of pefloxacin in actual samples

The lake water taken from Qinglian Lake in the south campus of Henan Polytechnic University was filtered through a filter membrane (0.22 μm), the suspended particles were removed to obtain the actual sample solution of the lake water, and different amounts of PFLX were added dropwise. Honey purchased from a nearby supermarket was diluted 20 times with ultrapure water and filtered with a 0.22 μm filter membrane to remove large particles. The treated supernatant was the actual sample solution of honey, and different amounts of PFLX were added dropwise to it for use. Soil sample (0.2 g) was taken from the south campus of Henan Polytechnic University, and was dissolved in 10 ML ultrapure water. After ultrasonic treatment for 30 min, the soil sample was centrifuged (9600 rpm/min, 10 min), filtered by 0.22 μm filter membrane, and the supernatant was stored for later use. The above Web-based concentration prediction system was used to detect PLFX.

3. Results and discussion

3.1. Characterization of BSA-AuNCs-GMP-Tb

The morphologies of BSA-AuNCs, GMP-Tb, and BSA-AuNCs-GMP-Tb were investigated by Transmission electron microscopy (TEM). Fig. 1(a) shows that BSA-AuNCs were spherical with an average particle size of about 28 nm, and aggregation occurred. Tb^{3+} reacted with the phosphate bond of the GMP molecule and the N atom in guanine base [27] to spontaneously form an irregular supramolecular polymer network (Fig. 1(b)). The average particle size of the colloidal spheres was about

22 nm. Fig. 1(c) shows the morphology of BSA-AuNCs-GMP-Tb, and it can be seen that GMP-Tb formed nano complexes with BSA-AuNCs, which may be due to the hydrogen bonding between BSA-AuNCs and GMP-Tb. The EDS spectrum of BSA-AuNCs-GMP-Tb (Fig. 1(d)) and the corresponding TEM-mapping imaging (Fig. 1(e–h)) illustrated that the presence and uniform distribution of Au, Tb and P elements indicated the successful preparation of BSA-AuNCs-GMP-Tb.

From the XRD data of GMP-Tb and BSA-AuNCs-GMP-Tb, as shown in Fig. 2(a), the composite of GMP-Tb and BSA-AuNCs did not show prominent diffraction peaks, indicating that the nanocomposite probe was amorphous structure [44]. In order to further analyze the structure of the nanocomposite probe, the elemental composition and surface groups were analyzed by Fourier transform infrared spectroscopy (FT-IR) and XPS scanning spectroscopy. As shown in Fig. 2(b), for BSA-AuNCs, the peaks at 1543 cm^{-1} , 1655 cm^{-1} , and 3332 cm^{-1} were caused by the bending vibration of N-H, the stretching vibration of C=O and the stretching vibration of N-H and O-H in BSA [26,45]. In the FT-IR spectrum of GMP-Tb, $620 - 725\text{ cm}^{-1}$ was the characteristic peak of the guanine base, and the peaks at 1008 cm^{-1} , 1076 cm^{-1} , 1635 cm^{-1} , and 1683 cm^{-1} were the symmetric stretching vibration, antisymmetric stretching vibration, C=O stretching vibration and P-OH stretching vibration of phosphate group [20,46,47]. Compared with the FT-IR spectra of GMP-Tb, the characteristic peak of the symmetric stretching vibration of the phosphate group in the FT-IR spectra of BSA-AuNCs-GMP-Tb shifted to 995 cm^{-1} , and the characteristic peak caused by the N-H bending vibration in the BSA-AuNCs complex appeared, indicating that BSA-AuNCs interacted with GMP-Tb to form the final coordination polymer.

The XPS full scan spectrum (Fig. S1) of the BSA-AuNCs-GMP-Tb probe showed six prominent peaks of Au 4f, P 2p, C 1s, N 1s, O 1s, and Tb 3d, indicating that the probe contained Au, P, C, N, O and Tb elements. The characteristic peaks at 284.85, 286.3, 287.6, and 288.2 eV in the high-resolution C 1s spectrum (Fig. S2(a)) corresponded to C-C/C=C, C-N/C-O, C=N and C=O bonds, respectively [48]. The characteristic peaks at 398.8, 399.88, and 400.7 eV in the N 1s energy spectrum (Fig. S2(b)) confirmed the presence of N = C, N-C, and C-N-H bonds. The characteristic peaks at 531.02, 531.46, 532.11, 532.54, and 533.13 eV in the O 1s spectrum (Fig. S2(c)) indicated O=P, O=C, O-C, O-P and C-OH bonds, respectively [49,50]. The characteristic peak at 133.42 eV in the P 2p energy spectrum (Fig. S2(d)) indicated the P-O/P = O bond [50]. In the Au 4f energy spectrum (Fig. S2(e)), the characteristic peaks at 84.25 eV and 88.1 eV represented Au 4f_{7/2} and Au 4f_{5/2}, respectively [28,51]. In the Tb 3d energy spectrum (Fig. S2(f)), the characteristic peaks at 1242.31 eV and 1276.91 eV binding energies corresponded to Tb³⁺ [52].

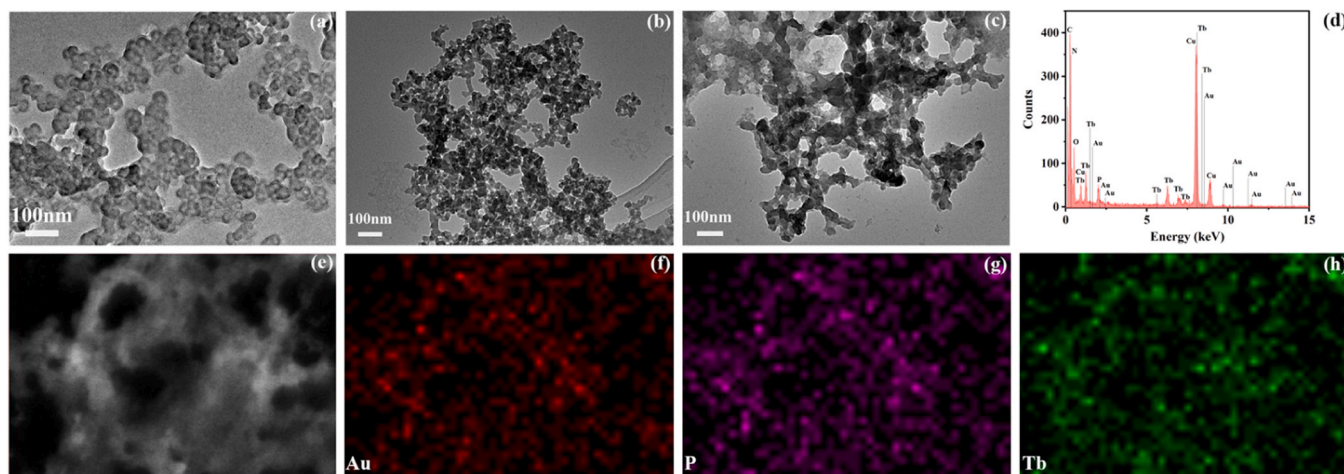


Fig. 1. TEM images of BSA-AuNCs (a), GMP-Tb (b), and BSA-AuNCs-GMP-Tb (c); EDS diagram of BSA-AuNCs-GMP-Tb (d); TEM-mapping of BSA-AuNCs-GMP-Tb (e–h).

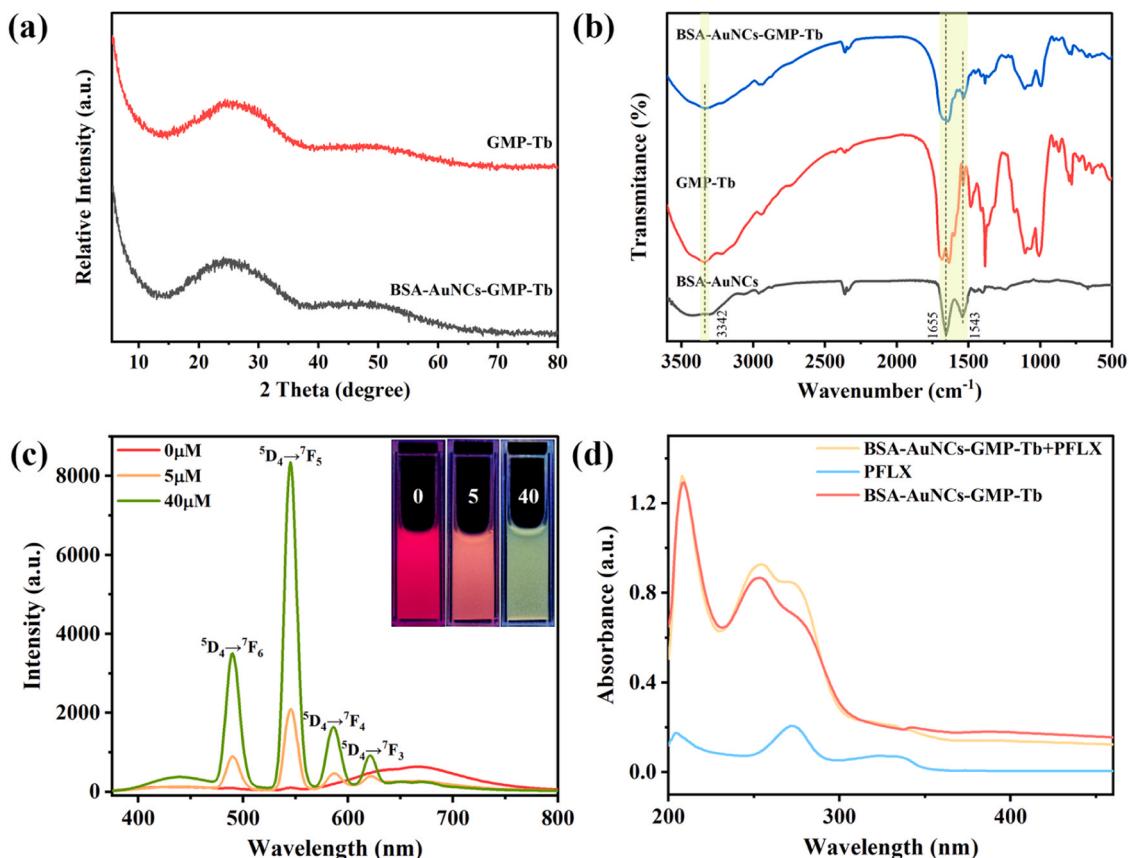


Fig. 2. (a) XRD patterns of GMP-Tb and BSA-AuNCs-GMP-Tb; (b) FT-IR spectra of BSA-AuNCs, GMP-Tb and BSA-AuNCs-GMP-Tb; (c) Fluorescence spectra of BSA-AuNCs-GMP-Tb treating 0 μM , 5 μM , and 40 μM PFLX; (d) UV absorption spectra of BSA-AuNCs-GMP-Tb before and after adding PFLX.

3.2. Sensing mechanism of fluorescent probe and optimization of detection conditions

3.2.1. Sensing mechanism

In this research, a multi-color fluorescence sensor (BSA-AuNCs-GMP-Tb) for the detection of PFLX was designed. The red fluorescence of BSA-AuNCs-GMP-Tb was generated by the emission peak of BSA-AuNCs at 669 nm. As shown in Fig. 2(c), when PFLX was not added, BSA-AuNCs-GMP-Tb did not show characteristic fluorescence at 545 nm. The possible reason was that Tb^{3+} had a non-radiative quenching effect with coordinated water molecules [53]. After adding 5 μM PFLX, PFLX containing α -keto acid structure chelated with Tb^{3+} , and the energy was transferred to Tb^{3+} through the 'antenna effect', which promoted the electron transition of Tb^{3+} from $^5\text{D}_4$ to $^7\text{F}_j$ ($j = 3, 4, 5, 6$) energy level [15], so that the emission spectra of the fluorescent probe showed the characteristic peaks of Tb^{3+} at 490 nm, 545 nm, 586 nm and 621 nm. Fig. S3(a) described the energy transfer mechanism from PFLX to Tb^{3+} . Based on the Latva's rule, the triplet state (T_1 , 22,487 cm^{-1}) [54] of PFLX can be directly transferred to the lowest excited state ($^5\text{D}_4$, 20,566 cm^{-1}) of Tb^{3+} [55]. When the electron returned from the excited state to the ground state, it may emit green fluorescence, of which the transition at 545 nm ($^5\text{D}_4 \rightarrow ^7\text{F}_5$) was the most sensitive. When 40 μM PFLX was added, PFLX continuously transferred the absorbed energy to Tb^{3+} through intramolecular energy transfer, sensitizing Tb^{3+} to emit green characteristic fluorescence, making the fluorescent probe exhibited green fluorescence under a 365 nm UV lamp. Fig. 2(d) displayed that the addition of PFLX did not cause a new absorption peak of BSA-AuNCs-GMP-Tb, and the intensity of the absorption peak at 272 nm increased, indicating PFLX and Tb^{3+} were chemically coordinated. The fluorescence lifetime of Tb^{3+} before and after the addition of PFLX was recorded in Fig. S3(c). These data were analyzed by the double

exponential function (Eq.(1)) in Origin 2021, and the corresponding average lifetime was calculated (Eq.(2)). Table S1 and Fig. S3(c) showed that the fluorescence lifetime of Tb^{3+} was prolonged from 0.501 ms to 0.659 ms after the addition of 5 μM PFLX. This may be due to the fact that PFLX replaced the water molecules at the unsaturated position of Tb^{3+} , which effectively inhibited the non-radiative quenching of water molecules and promoting the fluorescence enhancement of Tb^{3+} [56].

$$I = I_0 + A_1 \exp(-t/\tau_1) + A_2 \exp(-t/\tau_2) \quad (1)$$

$$\tau_{\text{av}} = (A_1 \tau_1^2 + A_2 \tau_2^2) / (A_1 \tau_1 + A_2 \tau_2) \quad (2)$$

Among them, I is the luminescence intensity, I_0 is the intensity when time (t) is 0, A_1 and A_2 are constants, t is time, and τ_1 and τ_2 are decay lifetime values.

The inner filter effect (IFE), fluorescence resonance energy transfer (FRET), dynamic quenching effect, static quenching effect and photo-induced electron transfer (PET) between the fluorescent probe and the detected substance can lead to the quenching of the fluorescence signal [57]. As shown in Fig. S3(b), the UV absorption spectrum of PFLX did not overlap with the emission spectrum of BSA-AuNCs-GMP-Tb fluorescent probe, but partially overlapped with its excitation spectrum, indicating that its quenching may be caused by IFE or FRET. The addition of PFLX reduced the average fluorescence lifetime from 5.269 μs to 5.134 μs (Fig. S3(d) and Table S2), indicating that the decay of the average lifetime after the interaction between BSA-AuNCs-GMP-Tb and PFLX was consistent with the energy transfer process between the fluorescent probe and PFLX [29], that was, FRET was the main mechanism for the fluorescence inhibition of the fluorescent probe at 669 nm after the addition of PFLX.

3.2.2. Optimization of detection conditions

Fig. S4(a) showed that in the detection range of excitation wavelengths from 310 nm to 360 nm, the fluorescence intensity of BSA-AuNCs was unchanged relative to the fluorescence intensity change of Tb^{3+} . While the characteristic peak at 545 nm of Tb^{3+} was the strongest excited on 335 nm, so 335 nm was chosen during the whole text. Since the amount of the probe may affect the color transition and initial fluorescence intensity of the fluorescence response, the change of the fluorescence intensity $I/I_{\max}$ ratio of the characteristic peak of Tb^{3+} at 545 nm in the range of 20 μL –100 μL ($C_{\text{PFLX}}=15 \mu\text{M}$) was explored. As shown in Fig. S4(b), the $I/I_{\max}$ ratio was maximum when the fluorescent probe dosage was 70 μL , and continued to decrease accompanied the increase of the fluorescent probe dosage. Hence, the fluorescent probe dosage was 70 μL in the whole experiment. As shown in Fig. S5(a), when the pH value was more significant than 4, the fluorescence intensity at 545 nm decreased with the increase of pH value. In comparison, the fluorescence intensity of BSA-AuNCs at 669 nm was relatively stable. Combined with the UV-vis spectra of PFLX in HEPES buffers with different pH (Fig. S5(b)), it was found that when the pH was more significant than 6, the UV-vis absorption peak of PFLX had a significant blue shift. The blue fluorescence under the UV lamp (Fig. S5(c)) gradually weakened to quenching, indicating that the structure of PFLX had changed. The optimal pH for the BSA-AuNCs-GMP-Tb nanofluorescent probe to detect PFLX was fixed at 4. Fig. S5(d) showed that the fluorescence intensity of the probe increased to more than 90% after 2 min of PFLX, indicating that the optimal reaction time was 2 min.

3.3. Optical and visual sensing

As shown in Fig. 3(a), the nanoprobe only showed a BSA-AuNCs emission peak at 669 nm without PFLX. When PFLX was added, the characteristic peaks of Tb^{3+} at 490 nm, 545 nm, 586 nm, and 621 nm were significantly enhanced due to the interaction between PFLX and Tb^{3+} and energy transfer. It can be seen from the diagram that the fluorescence intensity of the probe had been significantly improved by adding only 0.1 μM PFLX. When titrated to 5.4 μM , the intensity of the emission peak of BSA-AuNCs at 669 nm tended to stabilize, which suggested that the fluorescence color change of the BSA-AuNCs-GMP-Tb nanoprobe depended only on the antenna effect of Tb^{3+} . Fig. 3(b) shows that the maximum detection concentration of the fluorescent nanoprobe for PFLX was 60 μM . As shown in Fig. 3(c) and Fig. 3(d), the fluorescence intensity ratio of I_{545}/I_{669} had an excellent functional relationship with the concentration of PFLX (0–60 μM), especially a good linear relationship in the range of 0–10 μM ($Y=0.17089+1.51842 \times X$, $R^2=0.99837$). The minimum detection limit (LOD) of PFLX was calculated to be 5.37 nM based on a three-fold signal-to-noise ratio. The minimum detection limit (LOD) of PFLX was calculated to be 5.37 nM based on a triple-fold signal-to-noise ratio, which was much lower than the maximum residue limit of 5 $\mu\text{g}/\text{kg}$ (10.741 nM) of PFLX in honey specified in the Chinese national food safety standard GB 31650.1-2022 [58]. As shown in Table S3, compared with some detection materials, the BSA-AuNCs-GMP-Tb ratio fluorescent probe had a lower detection limit for PFLX, indicating that the probe had higher sensitivity. Although the CTAB-Cu-GR/CP material based on electrochemical method had a lower LOD for the detection of PFLX, the detection range was relatively narrow. In addition, based on the fluorescence emission spectrum data, the CIE coordinates (Table S4) in the sensing process of PFLX were calculated, and the corresponding CIE chromaticity diagram (Fig. 3(e)) was drawn according to these CIE coordinates. With the increase of the concentration of PFLX, the fluorescence color of the CIE chromaticity diagram showed a transition from red region to green region, consistent with the fluorescence color of the visual fluorescence images (Fig. 3(f)) taken under the 365 nm UV lamp. These rich visual color changes provided a broader space for the design and development of portable detection equipment.

3.4. Stability, selectivity, and immunity to interference

In order to test the photostability of BSA-AuNCs-GMP-Tb fluorescent nanoprobe, the fluorescence intensity data of the nanoprobe within 800 s under high-pressure xenon lamp irradiation before and after the addition of PFLX were investigated and normalized. Fig. S6 displayed that the fluorescence intensity of the two emission peaks at 545 nm and 669 nm was not affected by PFLX, indicating that the synthesized fluorescent nanoprobe had good light stability. Since the impurities in the actual sample may affect the detection effect, the selectivity and anti-interference of the BSA-AuNCs-GMP-Tb fluorescent nanoprobe were investigated under optimal experimental conditions. Firstly, different kinds of quinolone antibiotics were detected at the same concentration (25 μM), including ciprofloxacin, ofloxacin, norfloxacin, and pefloxacin. As shown in Fig. S7(a), BSA-AuNCs-GMP-Tb showed a solid response to PFLX only, and the fluorescence color under UV lamp (Fig. S7(c)) showed a yellowish-white color, which may be due to the hybridization of the red fluorescence of the Au nanoclusters, the characteristic green fluorescence of Tb^{3+} , and the blue fluorescence of PFLX. Secondly, the response effect of PFLX in the presence of common metal ions was tested, including cations (Mg^{2+} , Na^{+} , Fe^{3+} , Mn^{2+} , Cu^{2+} , Ca^{2+} , K^{+} , Pb^{2+} , Ag^{+} , Cd^{2+} , Al^{3+} , Co^{2+} , Zn^{2+} , and Ni^{2+}) and anions (Cl^{-} , SO_4^{2-} , NO_3^{-} , and $\text{CH}_3\text{COO}^{-}$). As shown in Fig. S7(b), the addition of metal ions did not affect the response signal of the fluorescent probe to PFLX. Fig. S7(d) showed the fluorescence visual color diagram of the fluorescent nanoprobe for the same concentration of PFLX (25 μM) and different metal ions under a 365 nm UV lamp, which proved that only in the presence of PFLX can the visual fluorescence color of the probe be induced to shift significantly. In summary, BSA-AuNCs-GMP-Tb can specifically detect PFLX and has good anti-interference ability, which provided a basis for the detection of PFLX in actual samples in the next step.

3.5. Paper-based visual detection

In order to realize the rapid visual detection of PFLX, a paper-based visual sensor with low price, simple operation, and convenience was designed. As shown in Fig. 3(g), the filter paper loaded with BSA-AuNCs-GMP-Tb fluorescent nanoprobe showed the red fluorescence of BSA-AuNCs under a 365 nm UV lamp. With the increase in PFLX concentration, the fluorescence color of the paper-based sensor changed from red to green. Combined with the Colour Recognition App in the smartphone, the R, G, and B channel color values of the fluorescence pictures of the paper-based sensor can be obtained. The fitting curve in Fig. 3(g) showed that there was a good linear relationship between PFLX concentration (X) and G/R value (Y) in the range of 0–10 μM ($Y=0.1837+0.6666 \times X$, $R^2=0.99346$), and a good functional relationship in the range of 0–60 μM ($Y=0.29998 \times X^{0.3911}$, $R^2=0.99332$).

3.6. Performance analysis of YOLO v5 model

In this research, the web-based PFLX concentration prediction system was developed using a deep learning-based YOLO v5 target detection algorithm for target detection and cropping. The lightweight version of YOLO v5s was selected to detect target cuvettes in images. Fig. 4 shows the model architecture of YOLO v5s [59]. The data used for the YOLO v5s model was derived from the fluorescent images taken during the daily experiment (Fig. S8). The camera and video function of the smartphone Glory 20 s took the video and fluorescent images. There were 950 images in the data set, divided into training set, verification set, and test set in the ratio of 0.56: 0.22: 0.22. The model training flow chart was shown in Fig. 5(a).

In order to verify the training effect of the model, the Precision (Eq. (3)), Recall (Eq. (4)), and mAP (Eqs. (5–6)) [60] were used to evaluate the model. The evaluation results were given in Table S5, which, combined with the Precision-Recall plot (Fig. 5(b)), illustrated that the

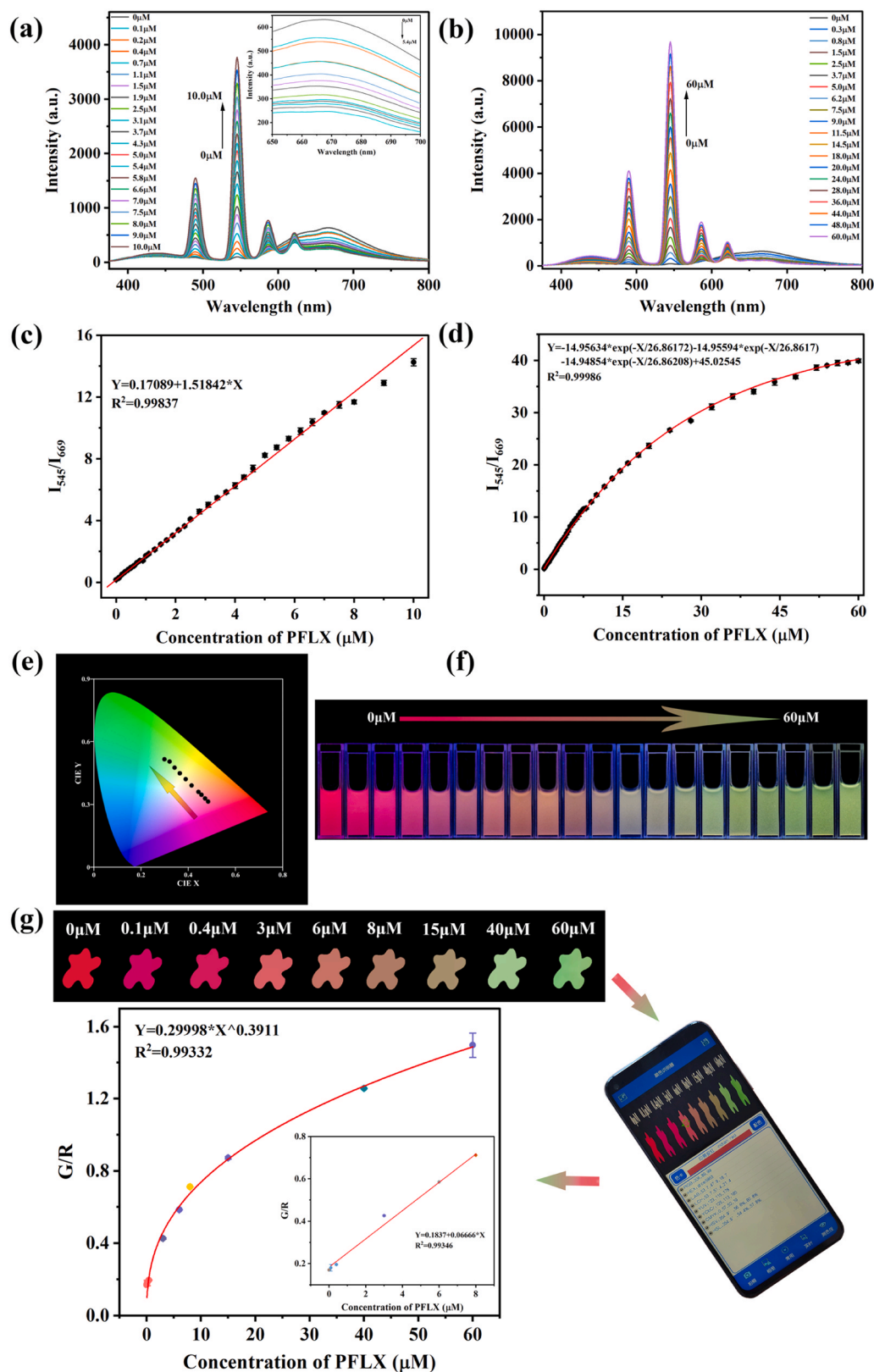


Fig. 3. Fluorescence intensity of BSA-AuNCs-GMP-Tb with PFLX concentration between 0 – 10 μM (a) and 0 – 60 μM (b); the linear relationship between the concentration of PFLX in the range of 0 – 10 μM (c) and 0 – 60 μM (d) and the fluorescence intensity ratio of I_{545}/I_{669} ; (e) CIE coordinates of BSA-AuNCs-GMP-Tb at different concentrations of PFLX; (f) The response fluorescence images of BSA-AuNCs-GMP-Tb to PFLX under 365 nm UV lamp; (g) Images of fluorescence changes in paper-based sensors after the addition of different amounts of PFLX and the fitted curves of chromaticity values and PFLX concentrations identified by the Colour Recognition App.

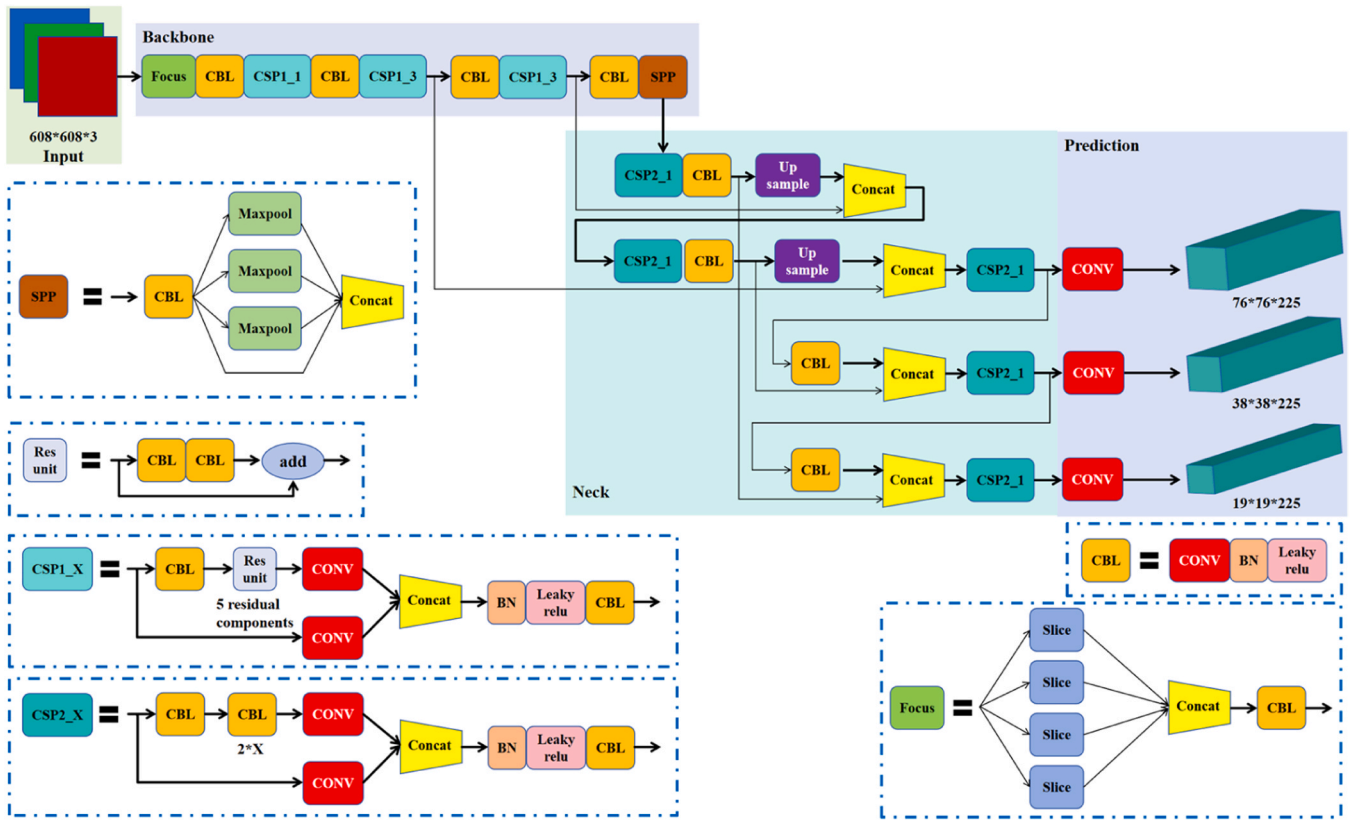


Fig. 4. The structure of YOLO v5s network.

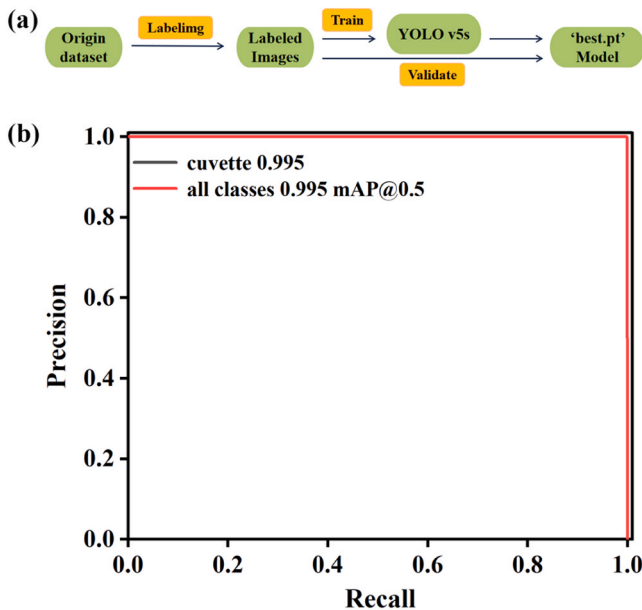


Fig. 5. (a) YOLO v5s model training flow chart; (b) Precision-Recall plot for the YOLO v5s training model.

YOLO v5s model had good learning results for the cuvette dataset.

$$\text{Precision} = \text{TP} / (\text{TP} + \text{FP}) \quad (3)$$

$$\text{Recall} = \text{TP} / (\text{TP} + \text{FN}) \quad (4)$$

$$\text{AP} = \int_0^1 P(r) dr \quad (5)$$

$$mAP = \frac{\sum_{i=1}^C (AP_i)}{C} \quad (6)$$

Among them, TP represents the number of predicted target cuvettes that are target cuvettes; FP represents the number of predicted target cuvettes that are not target cuvettes; FN denotes the number of unpredicted cuvettes. $P(r)$ indicates that the independent variable is the recall, and the dependent variable is a function of precision. The curve based on the $P(r)$ function is the Precision-Recall curve [61]. C denotes the number of predicted categories. When the predicted category is one, the value of AP is the value of mAP.

$$F1 = (2 * \text{Precision} * \text{Recall}) / (\text{Precision} + \text{Recall}) \quad (7)$$

In order to test the generalization ability of the final model, based on the test set, the Precision (Eq. (3)), Recall (Eq. (4)), and F1 (Eq. (7)) were used to evaluate the detection performance of the final model for cuvettes. The evaluation results shown in Table S6 indicated that the final model had good generalization ability and could accurately identify the cuvette in the picture.

3.7. Performance analysis of concentration prediction algorithms

This research used a variety of ML algorithms for model training, including linear regression (LR), K-nearest neighbor regression (KNN), support vector machine regression (SVR), decision tree regression (DT), extreme tree regression (ET), random forest regression (RF), ridge regression (RR), lasso regression (Lasso), multi-layer perceptron regression (MLP), gradient boosting (GBDT), adaptive boosting (Ada-Boost) and Bagging ensemble learning algorithm based on DT, ET and RF respectively.

Fig. 6(a) indicated that the green channel value had the highest correlation with the concentration of PFLX (0.84), which was consistent with the phenomenon that sensitization of Tb^{3+} by PFLX caused the

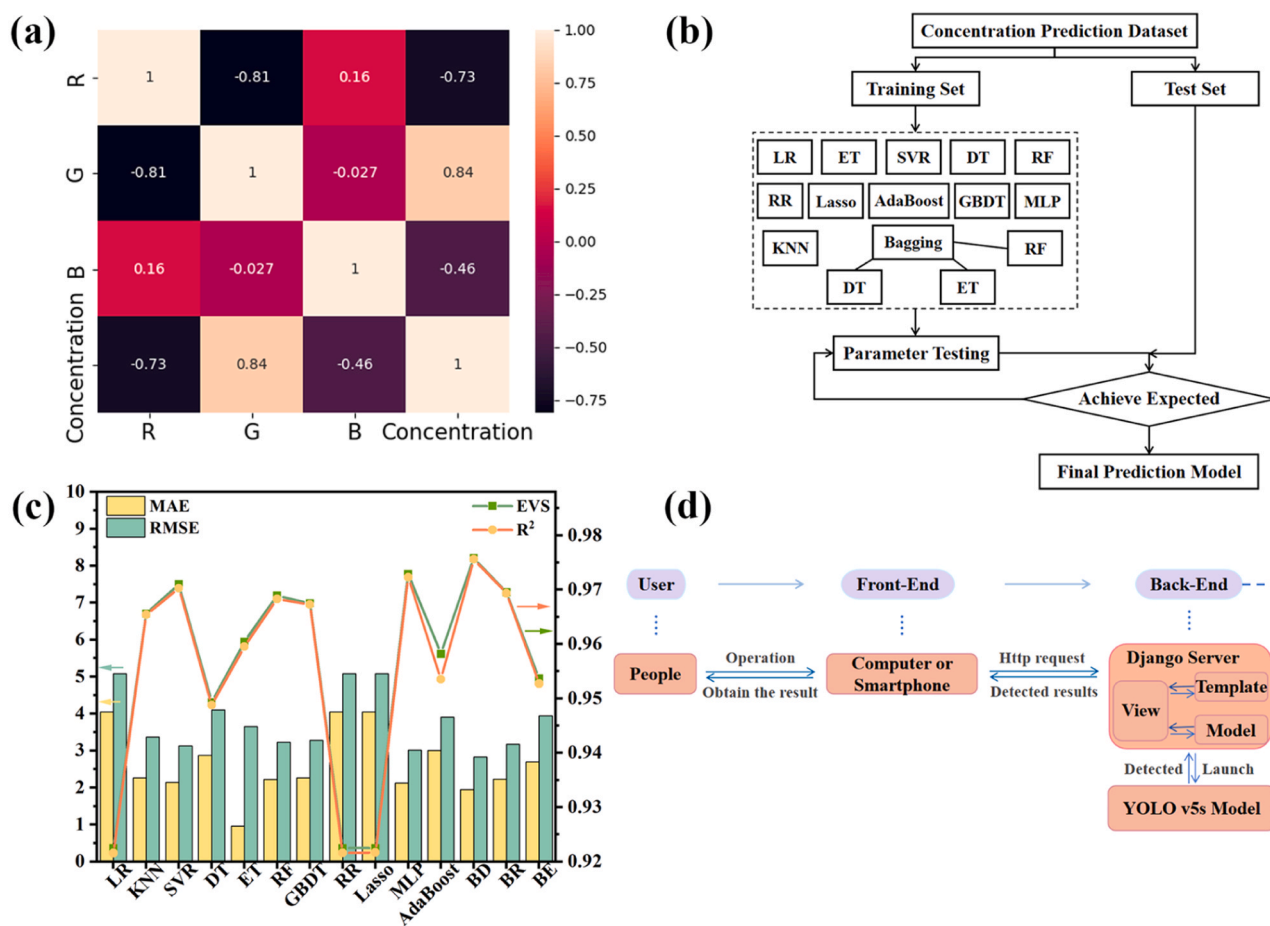


Fig. 6. (a) The heat map of R, G, and B three channels and PFLX concentration; (b) Flow chart of prediction model construction based on multiple ML algorithms; (c) Model evaluation graph based on multiple ML regression algorithms (BD, BR and BE in this graph represent Bagging ensemble learning algorithms based on DT, RF and ET respectively); (d) Work flow chart of concentration prediction system based on Web.

BSA-AuNCs-GMP-Tb fluorescent probe to show a dominant fluorescence of green. The dataset consisted of 478 data, divided into a training set and a test set in a ratio of 8:2. The process of building the prediction model was shown in Fig. 6(b).

To determine the learning effect of the model, this research employed the release variance score (EVS), the determination coefficient (R^2), the mean absolute error (MAE), and the root mean square error (RMSE) as the evaluation indicators. The comparison between the predicted value and the actual value of the model (Fig. S9) roughly reflected that the Bagging algorithm based on DT had good prediction performance. Combining the evaluation column-line diagram (Fig. 6(c)) and the evaluation results (Table S7), compared with other learners (training models), the DT-based Bagging ensemble learning algorithm had good learning ability. Under an excellent fitting degree (EVS:97.58%, R^2 :0.9756), the gap between the predicted and actual values was small (MAE:1.9417, RMSE:2.8317). In summary, the Bagging algorithm based on DT in this paper could find the law between the color value and the concentration more accurately, so the training model of this algorithm was embedded in the concentration prediction system for the prediction of fluorescent pictures uploaded by users.

3.8. Web-based pefloxacin prediction system

A certain amount of PFLX was added dropwise to a 2 ML stable system (70 μ L probe stock solution and 1930 μ L HEPES buffer (10 mM, pH=4)). After mixing well and placing under a 365 nm UV lamp, the fluorescent pictures or videos were captured using the camera function of a smartphone. The pictures taken or the screenshots in the video were

used as the pictures to be uploaded (the former was used in this part).

The system's concentration prediction workflow (Fig. 6(d)) mainly included three steps:

- (1) With the help of smart devices, users can access the prediction system welcome interface through the local development URL.
- (2) When the user uploaded a picture, the server was triggered to receive the picture, and the server view module (View) received the URL request, and rendered the upload in the front-end user interface in JSON format through the server template module (Template).
- (3) Similarly, the user clicked the prediction function button, sent a request to the Django server through the back-end, and called the YOLO v5s training model and the `c_predict()` function (customize the function used to construct the law between color value and concentration in the Django framework view module) to process and calculate the uploaded pictures. The final prediction results were transmitted to the server and visualized on the front-end user interface.

As shown in Fig. 7 and Video S1, the system can process a single picture or a batch of multiple pictures. In the "operation interface" interface, the user clicked the "Select File" button to select the picture, the image to be uploaded was displayed in the waiting list, and the "Start Uploading" button was clicked to upload the image. Layui's Table component displayed the successfully uploaded image information. Predicted the concentration of the picture by clicking the "Predict Concentration" button. The prediction results were shown in the

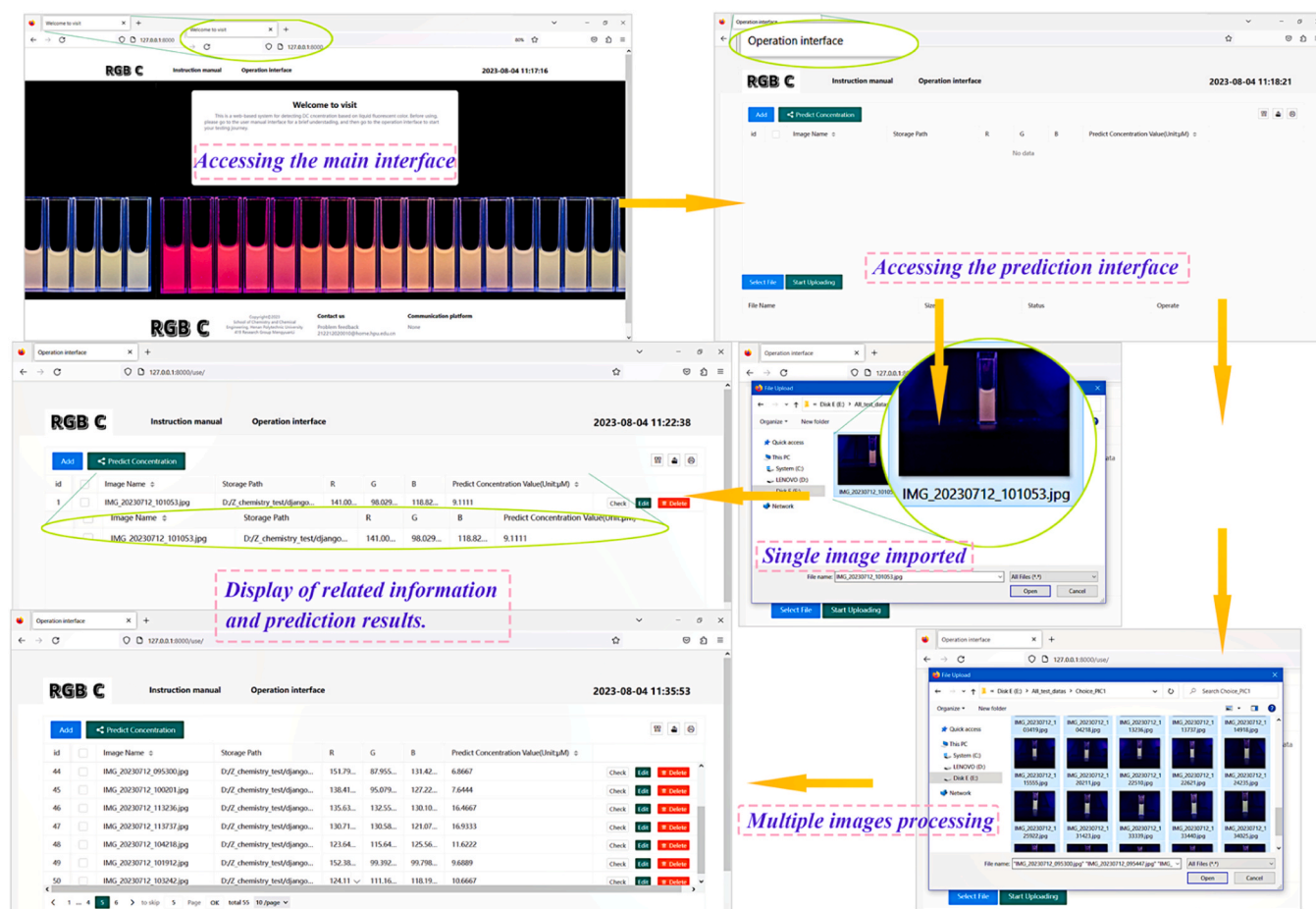


Fig. 7. The interface flow chart of a Web-based concentration prediction system for batch processing of single image and multiple images.

“Predict Concentration Value (Unit: μM)” column” of the Table component. As shown in Fig. S10, users can export data such as picture information and prediction results on demand, and the export format included CSV, Excel, and PDF.

Supplementary material related to this article can be found online at [doi:10.1016/j.snb.2024.135491](https://doi.org/10.1016/j.snb.2024.135491).

3.9. Detection of pefloxacin in lake water based on prediction platform

The sensing performance of BSA-AuNCs-GMP-Tb fluorescent nanoprobe was further applied to detect PFLX in actual lake water samples, honey samples and soil samples. In order to ensure the accuracy of the experimental results, the concentration data of each group were averaged three times. As shown in Table S8, the recovery of PFLX in lake water was between 96.8656% and 102.37%, and the maximum relative standard deviation (RSD) was 3.994%. The recovery of PFLX in honey was between 99.209% and 105.232%, and the maximum RSD was 4.408%. The recovery of PFLX in soil was 97.113%–102.381% and the maximum RSD was 2.349%. The test results showed that BSA-AuNCs-GMP-Tb was reliable and stable for detecting PFLX in lake water samples, honey samples and soil samples.

In addition, the fluorescence image of the sample solution ($\lambda_{\text{ex}}=365\text{ nm}$) was taken by a smartphone, and the PFLX in the simulated lake, honey and soil environment was detected using a Web-based concentration prediction system. The test results were shown in Table S9. The detection recovery of PFLX in lake water, honey and soil samples were 96.2%–102.2%, 95.407%–100.148% and 96%–108.647%, respectively, and the RSD was less than 7.378%. The test data showed that the Web-based concentration prediction system had a

good detection effect in practical environmental applications.

4. Conclusion

In brief, a BSA-AuNCs-GMP-Tb ratiometric nanoprobe with red fluorescence in the initial state was constructed by in situ preparation of guanine nucleotide-terbium ion complex using bovine serum albumin-coated gold nanoclusters as the substrate material. Under the optimal experimental conditions, based on different concentrations of PFLX, the interaction between the fluorescent probe and PFLX resulted in a naked-eye visual transition from red-orange-yellow to green under a 365 nm UV lamp. The colorful visual color provided a basis for the design of paper-based sensors. Combined with a wide range of detection range, a Web system based on YOLO v5 and a Bagging integrated learning algorithm was developed. The evaluation results illustrated that the final training model based on the YOLO v5 algorithm had good detection performance for the cuvette target, which guaranteed the extraction of cuvette color in the image. For the same data set, a variety of ML algorithms were trained. The Bagging ensemble algorithm based on DT had good learning ability and generalization ability, providing a reliable technical method for accurately detecting PFLX. At the same time, the use of the website eliminated the use of large instruments and the operation of professionals, which provided a broader visualization and prediction method for detecting PFLX or other antibiotics.

CRediT authorship contribution statement

Zhang Lina: Visualization, Software, Formal analysis. **Zhao Dan:** Validation, Methodology. **Xu Jun:** Writing – review & editing, Writing –

original draft, Supervision, Resources, Funding acquisition. **Zhao Tongqian**: Writing – review & editing, Supervision, Resources, Conceptualization. **Li Mengyuan**: Writing – original draft, Visualization, Validation, Software, Investigation, Formal analysis, Data curation. **Jia Lei**: Writing – review & editing, Writing – original draft, Resources, Project administration, Investigation, Funding acquisition, Data curation, Conceptualization. **Li Yongxin**: Writing – review & editing, Methodology, Investigation. **Chen Xiangzhen**: Visualization, Software, Formal analysis, Data curation.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

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Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at [doi:10.1016/j.snb.2024.135491](https://doi.org/10.1016/j.snb.2024.135491).

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